

The impact of large scale wind power generation on power system oscillations

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Abstract

Increasing amounts of wind turbines are connected to electrical power systems. This affects many aspects of their operation and behaviour. In this paper, the impact of large scale wind power generation on power system oscillations is treated. The three main types of power system oscillations, namely oscillations of a group of generators against a strong system and intra- and inter-area oscillations are studied. To this end, test systems are used. The effect of wind power on the oscillations is investigated by gradually replacing the power generated by the synchronous generators in the system by power from either constant or variable speed wind turbines, while observing the movement of the eigenvalues through the complex plane. From the results, it is concluded that the effect of wind power on power system oscillations depends on the wind turbine concept (constant or variable speed) and on the wind power penetration. In most cases, an increase in frequency and damping of power system oscillations was observed. This observation is caused by a relative reduction in size of the synchronous generators that engage in power system oscillations, which is equivalent to strengthening their mutual coupling. Further, it was shown that constant speed wind turbines damp power system oscillations more than variable speed turbines. This is caused by the damping effect of the squirrel cage induction generator used in constant speed wind turbines.

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1. Introduction

As a result of increasing environmental concern, the impact of conventional electricity generation on the environment is being minimised and efforts are made to generate electricity from renewable sources. The main advantages of electricity generation from renewable sources are the absence of harmful emissions and the in principle infinite availability of the prime mover that is converted into electricity. One way of generating electricity from renewable sources is to use wind turbines that convert the energy contained in flowing air into electricity. Up to this moment, the amount of wind power integrated into electrical power systems only covers a small part of the total power system load. The

rest of the load is for the largest part covered by conventional thermal, nuclear and hydro power plants.

As long as the wind power penetration level is low, the overall dynamic behaviour of the power system will mainly be determined by the synchronous generators in power plants. However, in some network areas, wind turbines gradually start to replace the output of conventional generators, especially during periods with low load and much wind. Because the characteristics of wind turbines are very different from those of conventional power plants, this development affects many aspects of power system operation and control, such as protection, frequency control, system balancing, transient stability, and voltage stability and control [1–3].

In this paper, a topic that is becoming increasingly important is investigated: the impact of wind power on electromechanical power system oscillations. Power system oscillations have been studied in many other

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contexts. Examples are the impact of HVDC links and their controllers on power system oscillations, the impact of long distance power transmission and the damping of power system oscillations by means of flexible AC transmission systems (FACTS) [4–6]. However, no studies in which the impact of wind power on power system oscillations is investigated have been reported.

The structure of this paper is as follows. First, the most important contemporary wind turbine concepts are described and the investigation of power system oscillations by eigenvalue analysis is introduced. Then, the test systems and the applied models are presented. The oscillation of a group of generators against a strong grid and intra-area and inter-area oscillations are investigated using test systems, in which the wind power penetration is gradually changed while observing the behaviour of the system's eigenvalues. To conclude, the results are clarified on the basis of existing theories on origin and mitigation of power system oscillations.

2. Research background

2.1. Wind power technology

The working principle of a wind turbine encompasses two main conversion processes. The turbine consists of a rotor that extracts kinetic energy from the wind and converts it into a rotating movement. This is then converted into electricity by a generator. Although this sounds rather simple, a wind turbine is a complex system in which knowledge from the areas of aerodynamics, mechanical, electrical and control engineering are applied [7].

When distinguished on the basis of the electromechanical conversion process, three different wind turbine concepts are currently mainly applied. The first is the constant speed wind turbine, which consists of a directly grid coupled squirrel cage induction generator. The wind turbine rotor is coupled to the generator through a gearbox. In most turbines of this type, the power extracted from the wind is limited using the stall effect. This means that the rotor is designed in such a way that its aerodynamic efficiency decreases in high wind speeds, thus preventing the mechanical power extracted from the wind from becoming too large. When the stall effect is used, no active control systems are necessary. However, pitch controlled constant speed wind turbines have also been built.

The second is a variable speed wind turbine with doubly fed (wound rotor) induction generator. The rotor winding is fed using a back-to-back voltage source converter. Like in the first concept, the wind turbine rotor is coupled to the generator through a gearbox. In

high wind speeds, the power extracted from the wind is limited by pitching the rotor blades.

The third concept is a variable speed wind turbine with a direct drive synchronous generator. The synchronous generator can have a wound rotor or be excited using permanent magnets. It is grid coupled through a back-to-back voltage source converter or a diode rectifier and voltage source converter. The synchronous generator is a low speed multi pole generator, therefore, no gearbox is needed. Like in the second concept, the power extracted from the wind is limited by pitching the rotor blades in high wind speeds. The concepts are depicted in Fig. 1.

2.2. Eigenvalue analysis

The behaviour of a dynamic system, of which an electrical power system is an example, can be described using the following general equation:

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{f}(\mathbf{x}, \mathbf{u}, t) \\ \mathbf{y} &= \mathbf{g}(\mathbf{x}, \mathbf{u})\end{aligned}\quad (1)$$

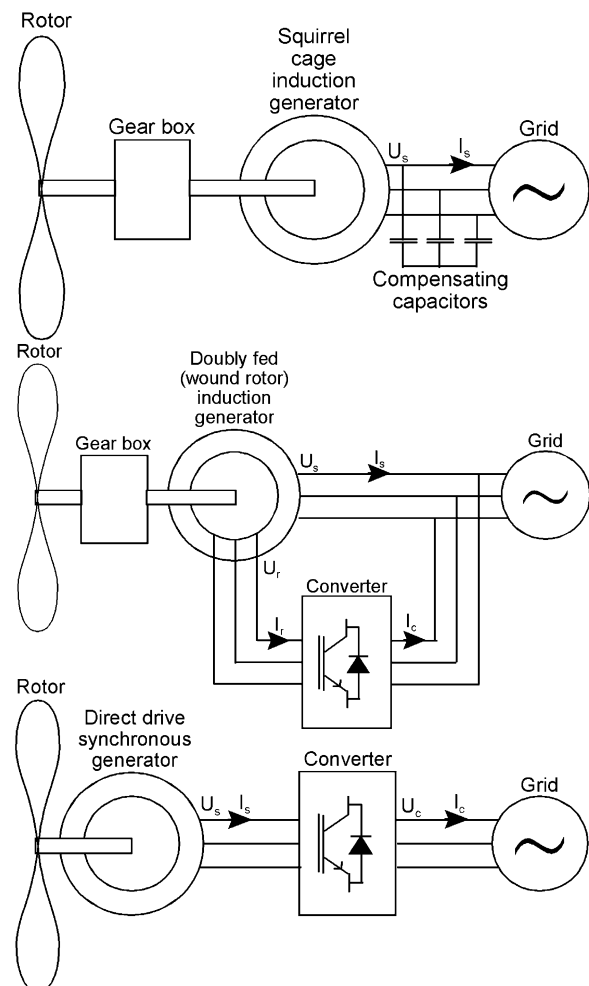


Fig. 1. Three most important contemporary wind turbine concepts.

$\mathbf{f}$ is a vector containing n first-order non-linear differential equations; $\mathbf{x}$ is a vector containing n state variables; $\mathbf{u}$ is a vector containing r input variables; $\mathbf{g}$ is a vector containing m non-linear algebraic equations; $\mathbf{y}$ is a vector containing m output variables; and t is time. By assuming that the system in (1) is time-invariant, i.e. the time-derivatives of the state variables are not explicit functions of the time t can be excluded from Eq. (1).

This system can be linearised and the resulting linearised description can be used to investigate the system's response to small variations in the input or state variables, starting at an equilibrium point [8]. To this end, Eq. (1) is expressed in terms of its Taylor's series expansion. With second and higher orders of the partial derivatives of $\mathbf{f}$ to the state variables omitted and only taking into account first-order terms, this gives the following for the i th component of vector $\mathbf{x}$:

$$\begin{aligned}\dot{x}_i &= \dot{x}_{i0} + \Delta \dot{x}_i = f_i(\mathbf{x}_0 + \Delta \mathbf{x}, \mathbf{u}_0 + \Delta \mathbf{u}) \\ &= f_i(\mathbf{x}_0, \mathbf{u}_0) + \frac{\partial f_i}{\partial x_1} \Delta x_1 \dots + \frac{\partial f_i}{\partial x_n} \Delta x_n + \frac{\partial f_i}{\partial u_1} \Delta u_1 \dots \\ &\quad + \frac{\partial f_i}{\partial u_r} \Delta u_r\end{aligned}\quad (2)$$

As can be seen from (1):

$$\dot{x}_{i0} = f_i(\mathbf{x}_0, \mathbf{u}_0) \quad (3)$$

and thus (2) can be written as:

$$\Delta \dot{x}_i = \frac{\partial f_i}{\partial x_1} \Delta x_1 \dots + \frac{\partial f_i}{\partial x_n} \Delta x_n + \frac{\partial f_i}{\partial u_1} \Delta u_1 \dots + \frac{\partial f_i}{\partial u_r} \Delta u_r \quad (4)$$

The same can be done for the j th component of $\mathbf{y}$:

$$\Delta \dot{y}_j = \frac{\partial g_j}{\partial x_1} \Delta x_1 \dots + \frac{\partial g_j}{\partial x_n} \Delta x_n + \frac{\partial g_j}{\partial u_1} \Delta u_1 \dots + \frac{\partial g_j}{\partial u_r} \Delta u_r \quad (5)$$

Doing this for all components of the vectors $\mathbf{x}$ and $\mathbf{y}$ gives the following linearised set of equations:

$$\begin{aligned}\Delta \dot{\mathbf{x}} &= \mathbf{A} \Delta \mathbf{x} + \mathbf{B} \Delta \mathbf{u} \\ \Delta \mathbf{y} &= \mathbf{C} \Delta \mathbf{x} + \mathbf{D} \Delta \mathbf{u}\end{aligned}\quad (6)$$

with

$$\mathbf{A} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \dots & \frac{\partial f_1}{\partial u_r} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \dots & \frac{\partial f_n}{\partial u_r} \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \dots & \frac{\partial g_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_m}{\partial x_1} & \dots & \frac{\partial g_m}{\partial x_n} \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} \frac{\partial g_1}{\partial u_1} & \dots & \frac{\partial g_1}{\partial u_r} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_m}{\partial u_1} & \dots & \frac{\partial g_m}{\partial u_r} \end{bmatrix} \quad (7)$$

Thus, the matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ contain the partial derivatives of the functions in $\mathbf{f}$ and $\mathbf{g}$ to the state variables $\mathbf{x}$ and the input variables $\mathbf{u}$.

Eq. (6) can be Laplace transformed. Thus, the state equations in the frequency domain are obtained:

$$\begin{aligned}s \Delta \mathbf{x}(s) - \Delta \mathbf{x}(0) &= \mathbf{A} \Delta \mathbf{x}(s) + \mathbf{B} \Delta \mathbf{u}(s) \\ \Delta \mathbf{y}(s) &= \mathbf{C} \Delta \mathbf{x}(s) + \mathbf{D} \Delta \mathbf{u}(s)\end{aligned}\quad (8)$$

A solution to the state equations can be obtained by rearranging the upper equation of (8) as follows:

$$(s\mathbf{I} - \mathbf{A}) \Delta \mathbf{x}(s) = \Delta \mathbf{x}(0) + \mathbf{B} \Delta \mathbf{u}(s) \quad (9)$$

The values of s which satisfy:

$$\det(s\mathbf{I} - \mathbf{A}) = 0 \quad (10)$$

are known as the eigenvalues of matrix $\mathbf{A}$ and Eq. (10) is defined as the characteristic equation of matrix $\mathbf{A}$. The eigenvalues of matrix $\mathbf{A}$ contain important information about the system's response to small perturbations of the state variable and its stability in the time domain.

In this research, the widely known power system dynamics simulation program PSS/E is used to calculate the eigenvalues of the investigated scenarios. In this program, a numerical algorithm is used to construct the matrix $\mathbf{A}$, which can then be used to calculate eigenvalues, left and right eigenvectors and participation factors, which together give information about any oscillatory modes that exist in the system and about the state variables that are involved in each of the system's modes [9]. Before applying PSS/E for this study, its eigenvalue calculation capabilities were thoroughly investigated by comparing the results obtained for small test systems to the literature and to the results yielded by analytical calculations and by another software package [10].

2.3. Power system oscillations

In general, three kinds of power system oscillations are distinguished namely [5,8,11]:

- Oscillations of one generator or a group of coherent generators against a strong system.
- Intra-area oscillations, i.e. oscillations of (groups of coherent) generators in a certain area of the network against each other, which, however, do hardly or not affect the rest of the generators in the network.
- Inter-area oscillations, i.e. oscillations of (groups of coherent) generators in a certain area of the network against (groups of coherent) generators in another

area of the network, also hardly affecting the rest of the generators in the network.

The various oscillation types can be distinguished on the basis of their mode shape. The shape of a mode that corresponds to a certain eigenvalue can be investigated using participation factors and eigenvectors. A participation factor indicates the relative contribution of each state variable to a certain mode. The elements in the right eigenvector belonging to a mode indicate the phase angle of the contribution of each of the state variables to that mode. High participation factors and differences in phase angle in the order of 180° indicate (groups of coherent) generators oscillating against each other. The location of these generators in the system determines the oscillation type. A group of generators that oscillates coherently will be further referred to as a swing node [4].

In general, the oscillation frequency becomes lower, starting from above in the list of oscillation types. Oscillations of the first type normally have a frequency above 1.0 Hz, of the second type between 0.4 and 0.7 Hz and of the third type between 0.1 and 0.3 Hz. However, in a large system with many oscillatory modes, this is not always true and the oscillation type should hence not be determined on the basis of the calculated oscillation frequency.

3. Research approach

3.1. Test system topologies

To investigate the impact of wind power on power system oscillations, two test systems that have been particularly developed for this study are used. The first system is depicted in Fig. 2. It consists of two areas, one with a large, strongly coupled system represented by an infinite bus and the other consisting of two synchronous generators. The impedances are in per unit on a 2500 MVA base and the loads are modelled as constant

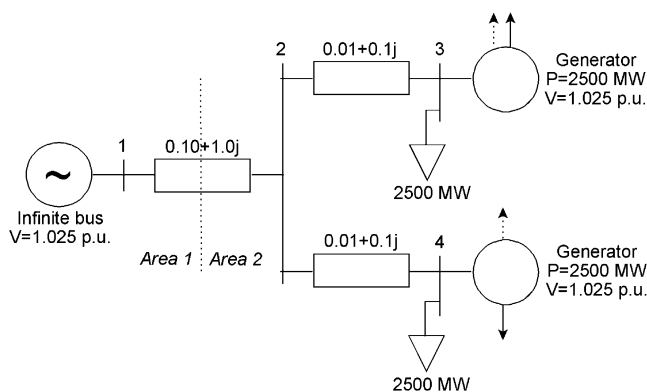


Fig. 2. Test system with two generators, the oscillatory modes are indicated with arrows: oscillation of a group of generators (dotted) and intra-area oscillation (solid).

MVA. This test system shows oscillations of the first two types: an oscillation of a group of generators and an intra-area oscillation. The generators at buses 3 and 4 oscillate against the strong system and generators 3 and 4 oscillate against each other. The shapes of these oscillatory modes are also depicted in Fig. 2.

The second test system is depicted in Fig. 3. The impedances are again in per unit on a 2500 MVA base and the loads are modelled as constant MVA. In this test system, the strongly coupled large system in area 2 of the first test system, is replaced by a system identical to that in area 1. Thus, no infinite bus is present in this test system. The test system shows oscillations of the second and third type. An inter-area oscillation between buses 1 and 2 and 5 and 6 and intra-area oscillations between buses 1 and 2 and between buses 5 and 6. The shapes of these oscillatory modes are also depicted in Fig. 3.

To investigate whether the obtained results were dependent on the loading of the tie line between the two areas, a power transfer between the two areas was introduced in both systems, by increasing each of the loads in area 2 of the systems with 500 MW (= 20%) and by reducing both loads in area 1 of the second system, depicted in Fig. 3, with 500 MW.

3.2. Applied models

To create dynamic models from the network topologies depicted in Figs. 1 and 2, a dynamic generator model must be attached to the generators in the load flow scenarios. In this study, a sixth order model of a round rotor generator with exponential modelling of magnetic saturation is used [9]. The generator model parameters can be found in Appendix A. In order to limit the number of states, to reduce the complexity of the problem and to improve the usability of the results, the synchronous generators are not equipped with governors and exciters. This is considered appropriate because the study aims at gathering qualitative insights. When accurate quantitative results from a real power system are pursued, a detailed representation of the governors and exciters of the generators would of course be very important.

As mentioned above and depicted in Fig. 1, there are different wind turbine types. The most important difference between constant speed wind turbines on the one hand and variable speed wind turbine types on the other is, that in a constant speed wind turbine a directly grid coupled generator is present, whereas in both variable speed wind turbine types, the electrical and mechanical behaviour of the generator are decoupled by the power electronic converter. Two different wind turbine models are used here, namely a constant speed wind turbine model and a general

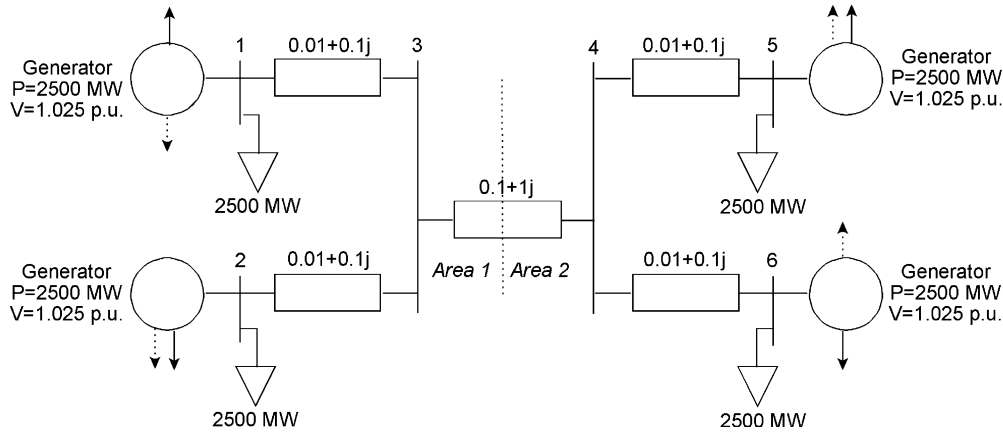


Fig. 3. Two area test system with four generators, the oscillatory modes are indicated with arrows: intra-area oscillation (solid) and inter-area oscillation (dotted).

variable speed wind turbine model that represents both variable speed wind turbine concepts.

The constant speed wind turbine is modelled as a third order (single cage) induction generator model, of which the parameters can be found in Appendix B. A two mass model is used for the mechanical part of the wind turbine. Wind turbines have relatively soft shafts. As a result, the eigenfrequency of the drive train is rather low and well within the bandwidth that is normally taken into account in power system dynamics simulations (0.1–10 Hz) and using a lumped mass representation for the mechanical part is not allowed [12]. For the variable speed wind turbine, a dedicated general variable speed wind turbine model was developed and integrated into PSS/E as a user defined model [9]. The general variable speed wind turbine model consists of:

- A wind speed model that generates a wind speed sequence.
- A rotor model, which calculates the mechanical power extracted from the wind.
- A model of the generator and converter, which are represented as a torque source, because in the frequency bandwidth that is of interest in power system dynamics simulations the power electronic converters are modelled as fundamental harmonic current sources, resulting in an algebraic relation between torque and current. This also allows modelling both variable speed wind turbine concepts with a single model in power system dynamics simulations [13].
- A model of the rotor speed controller, that determines the active power generation based on the actual value of the rotor speed.

In this study, it is not considered necessary to model all wind turbines in the wind park that (partially) replaces the synchronous generators individually. In-

stead, an aggregated wind park model, that interacts with the rest of the system in a way that is very similar to a detailed model, is used. This implies that the internal park network as well as any interactions between the turbines themselves are neglected. An aggregated model can, therefore, only be used when the subject of investigation is the impact of wind power on power system behaviour, as is the case here, and not when the design or behaviour of the wind park itself are of interest. The applied aggregation approach is different for constant and variable speed wind turbines and is described and verified elsewhere [14,15].

The aggregated wind park model is connected to the appropriate bus in the original system by an impedance of $0.015+0.15j$ p.u., as depicted in Fig. 4. The base value of this impedance equals the MVA rating of the wind park. Both constant and variable speed wind turbines are operated at a power factor equal to one. In case of constant speed wind turbines, this is achieved by applying compensating capacitors, whereas in case of variable speed wind turbines the power factor can be kept equal to one by controlling the power electronic converters appropriately [1]. The wind turbines do hence not contribute to voltage control.

3.3. Scenario development and calculations

Using the test systems depicted in Figs. 2 and 3, the load flow scenarios used to investigate the three oscillation types are developed in the following way:

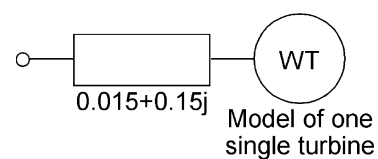


Fig. 4. Aggregated model, representing the wind park and its grid connection. The MVA base of the impedance is equal to the rating of the wind park.

- One or more buses to which a synchronous generator involved in the oscillatory mode to be studied is connected, are selected. The buses are selected according to the scenario being investigated. Both scenarios in which generators in only one of the two swing nodes are gradually replaced by wind power and scenarios in which generators in both swing nodes are gradually replaced by wind power are investigated.
- An aggregated model of a wind park with either constant or variable speed wind turbines, whose structure is depicted in Fig. 4, is connected to the selected buses.
- The active power, reactive power capability and MVA rating of the selected synchronous generator(s) are gradually reduced. The reduction in active power is compensated by increasing the power generated by the wind park, whose MVA rating is increased accordingly. The reduction in reactive power generation, if any, is not compensated. The impedance between the wind park and the grid is also changed according to the MVA rating of the wind park, because it is constant on the wind park's MVA base as can be seen in Fig. 4, but not on the network's MVA base.

In order to calculate the eigenvalues for each of the scenarios, the load flow cases are solved and dynamic models of the synchronous generators and the wind park(s) are attached. The resulting dynamic model of the investigated scenario is then linearised and the eigenvalues of the state matrix are calculated as described above. In this way, the trajectory of the eigenvalues in the complex plane with changing wind power penetration can be depicted, which gives information on the oscillatory behaviour. The above sequence of activities has been carried out both for the cases without and with an inter-area power transfer. In total, the eigenvalues of 164 scenarios have been calculated.

4. Effect of wind power on oscillations

4.1. Oscillation against a strong system

The oscillation of the generators at buses 3 and 4 against the infinite bus, as depicted in Fig. 2, is taken as an example of an oscillation of a group of generators against a strong system. The wind power penetration level is changed stepwise from zero in the base case to 50% eventually. First, it is changed in steps of 250 MW at bus 3 until the synchronous generator is fully replaced by the wind park. Then, it is changed in steps of 125 MW at buses 3 and 4 until half of the power of each of the synchronous generators (= 1250 MW) is replaced by wind power. In this way, it can be investigated whether

the effect of the wind power on the oscillation depends on the way in which the wind power is spread throughout the swing node, that here consists of the generators at buses 3 and 4. This is done for both constant and variable speed wind turbines and for scenarios without and with an inter-area power transfer. Thus, 82 eigenvalues are obtained, namely the base case either without and with inter-area power transfer and then ten values for the scenarios with different wind power penetrations, different wind turbine technologies and different wind park locations.

The results are depicted in Fig. 5. The damping ratio is indicated on the horizontal axis and the oscillation frequency on the vertical axis. The upper figure depicts the eigenvalues that correspond to one wind park, at bus 3. The lower figure depicts the eigenvalues that correspond to two wind parks, at buses 3 and 4. The meaning of the symbols is given the figure's legend. The direction in which the eigenvalues move with increasing wind power penetration is indicated with an arrow.

4.2. Intra- and inter-area oscillations

The system depicted in Fig. 2 shows an intra-area oscillation between buses 3 and 4. Again, the amount of wind power is changed till the penetration is 50%, first in 250 MW steps at bus 3 and then in 125 MW steps at buses 3 and 4, both for the system without and with an inter-area flow. This corresponds to erecting wind power at one or both of the swing nodes involved in the oscillation, respectively. The results are depicted in Fig. 6.

The system in Fig. 3 shows both an intra-area and an inter-area oscillation. The system is analysed by changing the wind power penetration at buses 1 and 2 and at buses 1, 2, 6 and 7 in steps of 250 and 125 MW, respectively, until the synchronous generators at buses 1 and 2 are completely replaced, or all synchronous generators are replaced for 50% (= 1250 MW). In the first case, both swing nodes of one of the two intra-area oscillations are affected and the swing nodes involved in the other intra-area oscillation are not affected. In the second case both swing nodes of each of the two intra-area oscillations are affected. With respect to the inter-area oscillation, in the first case only one swing node is affected, whereas in the second case both swing nodes are affected.

In Fig. 7, the results for the two intra-area oscillations that are present in the system depicted in Fig. 3 are shown. In Fig. 8, the results for the inter-area oscillation are shown. Results are given for the system without and with an inter-area power flow.

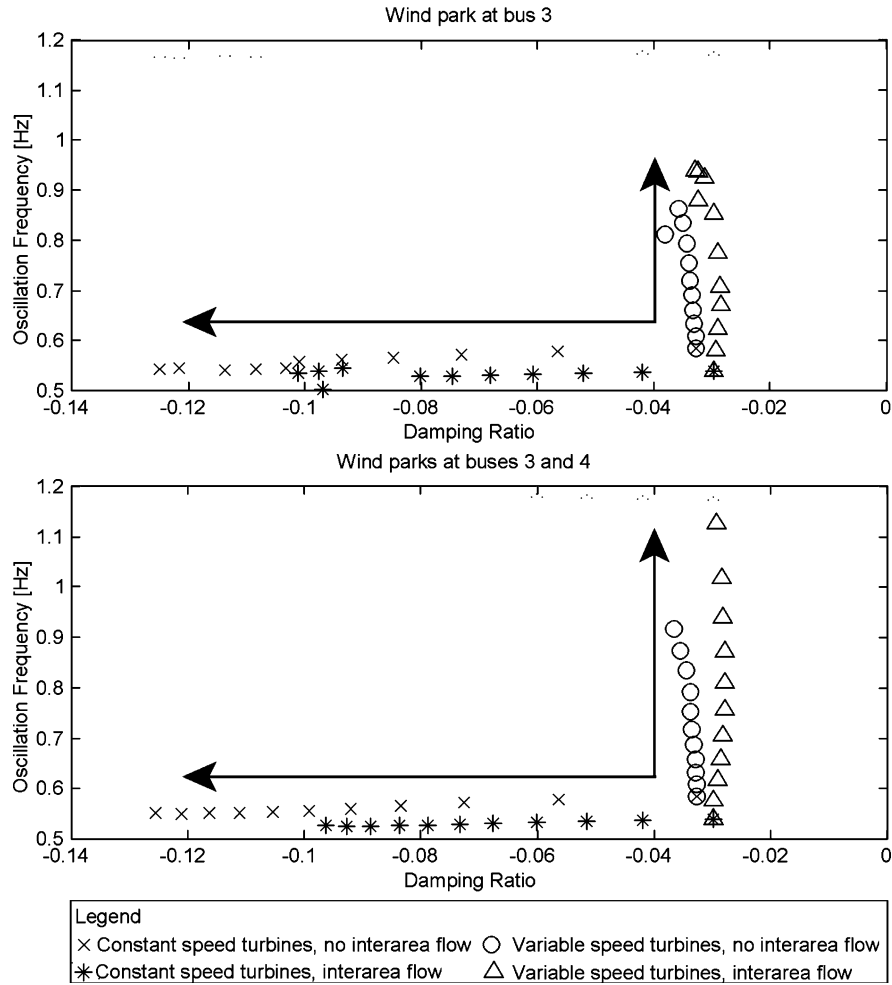


Fig. 5. Impact of increasing wind power penetration on the oscillation of the generators at buses 3 and 4 against the infinite bus. The upper figure depicts the eigenvalues with a wind park at bus 3, the lower figure the eigenvalues with two wind parks, at buses 3 and 4. The meaning of the symbols is indicated in the legend. The direction of the arrow indicates increasing wind power penetration.

4.3. Analysis of results

From the results depicted in Figs. 5–8, the following general conclusions can be drawn:

- Replacement of synchronous generators by wind power mainly affects oscillations of a (group of coherent) generator(s) against a strong system (see Fig. 5) and to a lesser extent also inter-area oscillations (see Fig. 8). The impact on intra-area oscillations is not significant (see Figs. 6 and 7).
- Constant speed wind turbines increase the damping of power system oscillations in all cases, whereas their impact on the frequency varies, depending on the type of oscillation.
- Variable speed wind turbines increase the frequency of power system oscillations in all cases, whereas their impact on the damping is rather limited and varies, depending on the type of oscillation.
- Replacement of synchronous generators by wind power does not significantly affect oscillations whose

swing nodes are not influenced by the substitution of synchronous generation by wind power (see second graph of Fig. 7).

- Although the impact of wind power on power system oscillations is affected by the loading of tie lines in a quantitative sense, its impact does not change qualitatively when tie line loadings are varied.

These findings can be explained by considering the physical origin of power system oscillations. In synchronous generators, the electrical torque is mainly dependent on the angle between rotor and stator flux. This angle is the integral of the rotational speed difference between these two fluxes, which is in turn dependent on the difference between electrical and mechanical torque. This makes the mechanical part of the synchronous machine a second-order system that shows an intrinsic oscillatory behaviour. Further, small changes in rotor speed do hardly affect the electrical torque developed by the machine, as they hardly lead to a change in rotor angle. Therefore, the damping of rotor speed oscillations

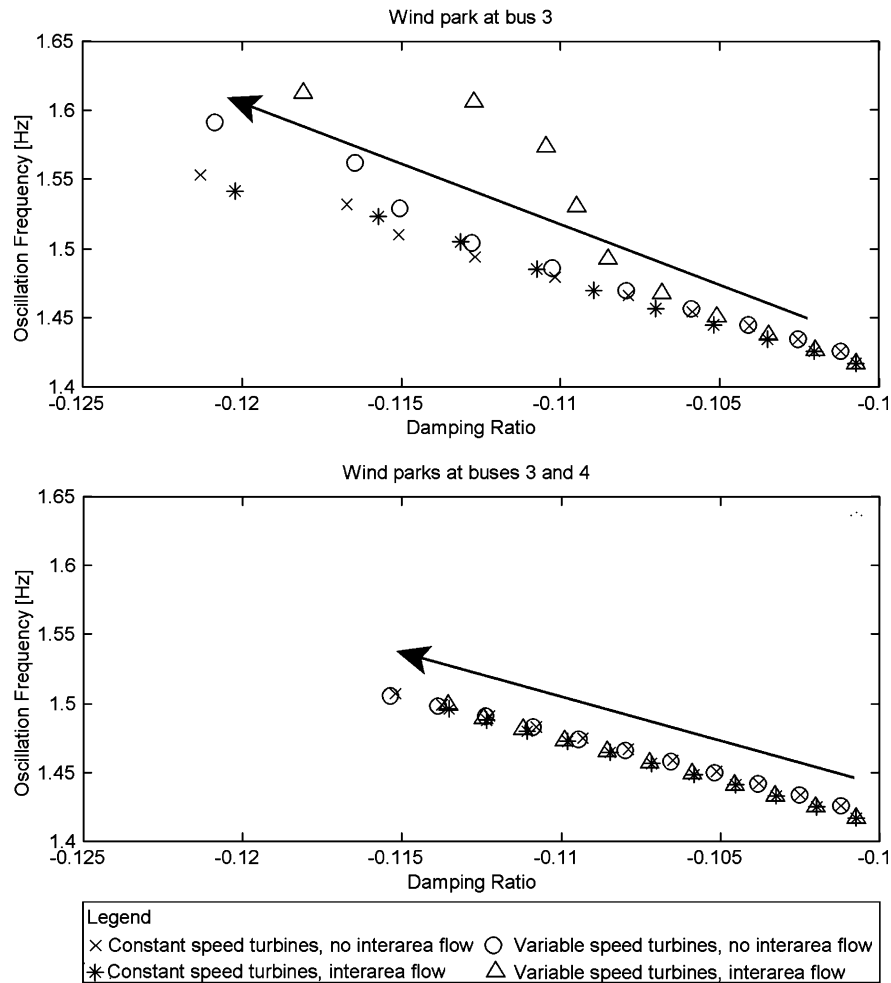


Fig. 6. Impact of increasing wind power penetration on the intra-area oscillation of the generators at buses 3 and 4 against each other. The upper figure depicts the eigenvalues with a wind park at bus 3, the lower figure the eigenvalues with two wind parks, at buses 3 and 4. The meaning of the symbols is indicated in the legend. The direction of the arrow indicates increasing wind power penetration.

must come from other sources, such as damper windings, the exciter and the rest of the power system.

Relatively weak links and large concentrations of synchronous generators are known to contribute to the risk of weakly damped or undamped oscillations [4,16]. This can be explained by noticing that when a synchronous generator is large compared with the scale of the system as a whole and/or when it is weakly coupled, the contribution of the rest of the system to the damping torque becomes less and thus the damping of oscillations becomes worse. The oscillation of a generator that is large compared with the system will also affect other generators, thus spreading the oscillation through the system and giving rise to inter and intra-area oscillations in which a number of generators takes part and oscillate

against each other. The lower the frequency, the less damping is provided by the damper windings. Because intra- and inter-area oscillations have frequencies of about 1 Hz and lower, hardly any damping is provided by the damper windings.

In the above investigations, it was assumed that wind power replaces the power generated by synchronous generators. Thus, the contribution of synchronous generators to the demand for power becomes less. The topology of the system stays, however, unchanged. Thus, the synchronous generators become smaller relatively to the impedances of the grid. This strengthens the mutual coupling, which in most cases improves the damping of any oscillations that occur between the synchronous generators.

Fig. 7. Impact of increasing wind power penetration on the intra-area oscillations of the generators at buses 3 and 4 and at buses 5 and 6 against each other. From above: wind power at buses 1 and 2, oscillation between generators at buses 1 and 2; wind power at buses 1 and 2, oscillation between generators at buses 5 and 6; wind power at buses 1, 2, 5 and 6, oscillation between generators at buses 1 and 2; wind power at buses 1, 2, 5 and 6, oscillation between generators at buses 5 and 6. The meaning of the symbols is indicated in the legend. The direction of the arrow indicates increasing wind power penetration.

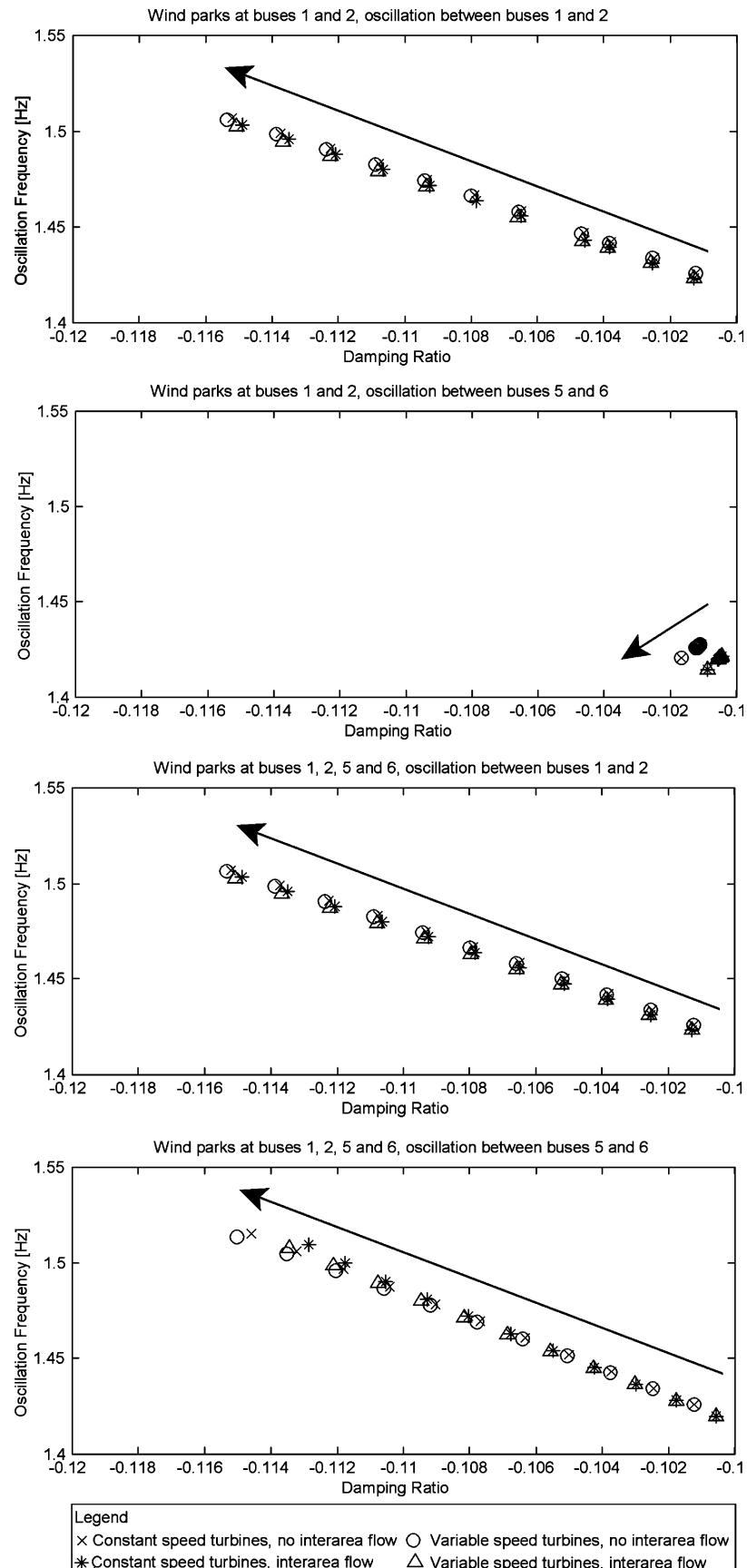


Fig. 7.

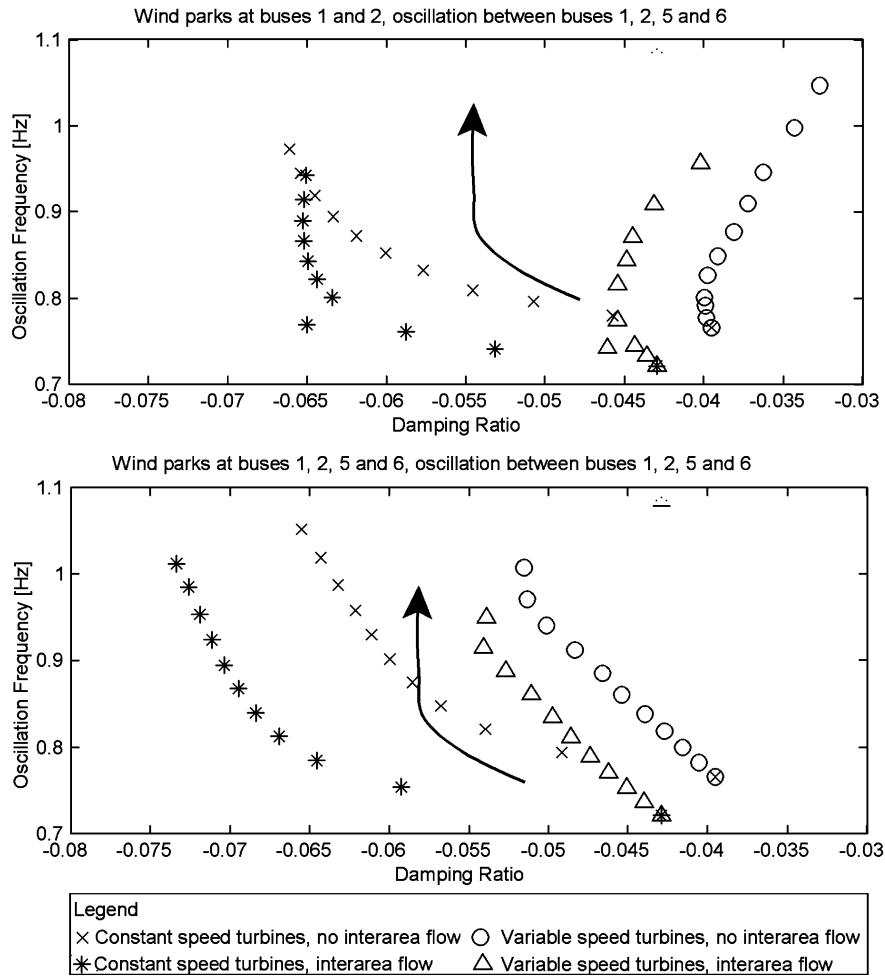


Fig. 8. Impact of increasing wind power penetration on the inter-area oscillation of the generators at buses 1 and 2 and 5 and 6 against each other. The upper figure depicts two wind parks, at buses 1 and 2, the lower figure four wind parks, at buses 1, 2, 5 and 6. The meaning of the symbols is indicated in the legend. The direction of the arrow indicates increasing wind power penetration.

The generator concepts used in wind turbines do not engage in power system oscillations. The squirrel cage induction generator used in constant speed wind turbines, shows a relation between rotor slip, i.e. rotor speed, and electrical torque, instead of between rotor angle and electrical torque. The mechanical part is, therefore, of first-order and does not show oscillatory behaviour, as opposed to that of a synchronous generator. Although an oscillation can be noticed when including the rotor transients in the model, because this increases the model order, this oscillation is small and well damped. Thus, squirrel cage induction generators are intrinsically better damped and rely less than synchronous generators on the power system to provide damping, and do thus hardly lead to power system oscillations.

The generator concepts used in variable speed wind turbines are decoupled from the power system by power electronic converters that control the rotor speed and electrical power and damp any rotor speed oscillations that may occur. Variable speed wind turbines do also

not react to any oscillations that may exist in the power system, because the generator does not notice those as they are not transferred through the converter.

The differences between the impact of constant and variable speed wind turbines that were observed can also be explained from the difference in working principles between the two concepts. A squirrel cage induction generator tends to damp power system oscillations [17]. This is due to the fact that an increase in synchronous generator speed would lead to a slight voltage increase. This shifts the rotor speed versus power curve of the squirrel cage induction generator, which has a damping effect on the power system oscillation. This effect does not occur in case of variable speed wind turbines, in which the generator is decoupled from the grid by the power electronic converter. This explains why constant speed wind turbines affect the damping of power system oscillations more than variable speed wind turbines.

It was also observed that when the synchronous generating capacity at one of the swing nodes involved in an oscillation is replaced by wind power to a large

extent, the impact of wind power becomes ambiguous, particularly in the case of inter-area oscillations (see upper graphs of Figs. 6 and 8). This is probably caused by the fact that in this situation, the nature of the oscillation changes. The change of the nature of the oscillation was observed when studying the mode shape using the eigenvectors corresponding to the relevant eigenvalue. By reducing the rating of the synchronous generators in one of the swing nodes of the oscillation, it becomes smaller when compared with the other swing node. This results in a change in mode shape and in some cases also of oscillation type.

5. Conclusions and discussion

In this paper, the impact of wind power on power system oscillations was investigated. This was done by observing the behaviour of the eigenvalues when changing the wind power penetration in two test systems. The test systems were developed to show all three distinct types of power system oscillations. To investigate the impact of the tie line loading, both cases with an unloaded and a loaded tie line were investigated.

It was concluded that wind power tends to improve the damping of oscillations of a (group of coherent) generator(s) against a strong system and of inter-area oscillations, particularly when constant speed wind turbines are used. The impact on intra-area oscillations is not significant. The damping effect of wind power can be explained by noticing that the remaining synchronous generators become relatively smaller and thus more strongly coupled. On the other hand, wind power itself does not induce new oscillatory modes, because the generator concepts used in wind turbines do not engage in power system oscillations: oscillations in a squirrel cage induction generator, that is used in constant speed wind turbines, are intrinsically better damped and the generators of variable speed wind turbines are decoupled from the power system by a power electronic converter, which controls the power flow and prevents them from engaging in power system oscillations.

When a very large part of the synchronous generation capacity in a swing node is replaced by wind power, the results become ambiguous, which is probably caused by the fact that the mode shape changes, which can also change the oscillation type.

In this paper, only two relatively small test systems were investigated. Because the observations could be explained using theoretical insights with respect to the origin and the mitigation of power system oscillations, it can be expected that investigations on other power systems will yield similar results. The results presented in this paper are, however, mainly qualitative. In order to investigate whether it is possible to quantify the impact

of wind power on power system oscillations, computations with other power systems would be necessary.

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Appendix A: Synchronous generator model parameters, MVA base equals 1.25 times active power

Quantity	Value	Quantity	Value
T'_{d0}	5.0 s	X_d	1.65 p.u.
T''_{d0}	0.05 s	X'_d	0.30 p.u.
T'_{q0}	1.0 s	X'_q	0.75 p.u.
T''_{q0}	0.04 s	$X''_d = X''_q$	0.20 p.u.
H	4 s	X_l	0.175 p.u.
D	0	$S(1.0)$	0.2
X_d	1.75 p.u.	$S(1.2)$	0.4

* X''_d equals X''_q due to the PSS/E synchronous generator model structure.

Appendix B: Asynchronous generator model parameters, MVA base equals 1.25 times active power

Quantity	Value	Quantity	Value
T'	5.0 s	X	3.1 p.u.
T''	0.05 s	X'	0.178 p.u.
H_{gen}	0.5 s	X_l	0.0 p.u.
H_{turbine}	2.5 s	$S(1.0)^*$	0.0
Shaft stiffness	0.3 p.u. torque/el. rad	$S(1.2)^*$	0.0

* Magnetic saturation is neglected.

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